

A HIGH INTENSITY INTERVAL TRAINING (HIIT)-BASED RUNNING PLAN IMPROVES ATHLETIC PERFORMANCE BY IMPROVING MUSCLE POWER

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ABSTRACT

García-Pinillos, F, Cámara-Pérez, JC, Soto-Hermoso, VM, and Latorre-Román, PÁ. A High Intensity Interval Training (HIIT)-based running plan improves athletic performance by improving muscle power. *J Strength Cond Res* 31(1): 146–153, 2017—This study aimed to examine the effect of a 5-week high-intensity intermittent training (HIIT)-based running plan on athletic performance and to compare the physiological and neuromuscular responses during a sprint-distance triathlon before and after the HIIT period. Thirteen triathletes were matched into 2 groups: the experimental group (EG) and the control group (CG). The CG was asked to maintain their normal training routines, whereas the EG maintained only their swimming and cycling routines and modified their running routine. Participants completed a sprint-distance triathlon before (pretest) and after (posttest) the intervention period. In both pretest and posttest, the participants performed 4 jumping tests: before the race (baseline), postswim, postcycling, and postrun. Additionally, heart rate was monitored (HRmean), whereas rate of perceived exertion (RPE) and blood lactate accumulation (BLa) were registered after the race. No significant differences ($p \geq 0.05$) between groups were found before HIIT intervention (at pretest). Significant group-by-training interactions were found in vertical jumping ability and athletic performance: the EG improved jumping performance ($\sim 6\text{--}9\%$, $p \leq 0.05$, effect size (ES) > 0.7), swimming performance ($p = 0.013$, ES = 0.438), and running time ($p = 0.001$, ES = 0.667) during the competition, whereas the CG remained unchanged ($p \geq 0.05$, ES < 0.4). No changes ($p \geq 0.05$, ES < 0.4) were observed in RPE, HRmean, and BLa. A linear regression analysis showed that ΔCMJ predicted both the $\Delta\text{Ru_time}$ ($R^2 = 0.559$; $p = 0.008$) and the $\Delta\text{Overall_time}$ ($R^2 = 0.391$; $p = 0.048$). This low-volume, HIIT-based running plan combined with the high training volumes of these triathletes in

swimming and cycling improved athletic performance during a sprint-distance triathlon. This improvement may be due to improved neuromuscular characteristics that were transferred into improved muscle power and work economy.

KEY WORDS endurance athletes, interval training, muscular performance parameters, training prescription

INTRODUCTION

Increased participation in recreational and competitive triathlons over the last decade has been accompanied by an increase in the number of athletes sustaining injuries (9). Studies investigating factors relating to levels of training that contribute to injury identified that training for or competing in the running component of the triathlon resulted in the greatest number of, and often the most severe, injuries (9,28). Specifically, risk of injury increased with increased weekly training distances, especially for running (9,28). In contrast, a growing body of literature points to mean training intensity over a season as the key factor for performance improvement (8,20). A clear example for endurance sports was reported by Billat et al. (4) who showed that male Kenyan runners training at higher speeds had a significantly better 10-km performance than Kenyan athletes training at lower speeds, despite the elite status of both groups.

As a training method that leads to a reduction in weekly running distances and an increase in mean running intensity without impairing athletic performance, high-intensity interval training (HIIT) is considered one of the most effective forms of exercise for improving the physical performance of athletes (2,3,7,8,20). Although there is no universal definition, HIIT generally refers to repeated short to long bouts of high-intensity exercise—performed at close to 100% maximal oxygen uptake ($\dot{V}\text{O}_2\text{max}$)—interspersed with recovery periods.

The HIIT protocol is well documented (8), and various types of HIIT programs have been shown to improve endurance performance in runners (1,12,17) and cyclists (10,19). However, despite the reported benefits of training at a high intensity, endurance athletes continue to train mostly at low intensities (13); thus, more evidence is needed to “convince” coaches and

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athletes of the importance of HIIT for endurance performance. Finally, all of these studies have been performed in a single sport, by replacing a part of their training programs and reducing, in a variable percentage (0–50%), the average training distance. Nevertheless, to date, no previous studies have proposed any strategy to insert and apply the HIIT methodology to a triathlon.

Taken together, the aim of this study was to examine the effect of a 5-week HIIT-based running plan on the athletic performance of triathletes and to compare the physiological and neuromuscular responses during a sprint-distance triathlon race before and after this high-intensity and low-volume intervention period. The authors hypothesize that a low-volume HIIT-based running plan, combined with the already high training volumes of these triathletes in swimming and cycling, might be a more efficient training program for improving the performance of triathletes than the typically performed high-volume and low-/moderate-intensity exercise.

METHODS

Experimental Approach to the Problem

This study analyses the effect of incorporating HIIT on muscle power measurement and simulated sprint triathlon performance. Using a between-group design (experimental group [EG] and control group [CG]), 13 triathletes were assessed. Athletes from the CG were asked to maintain their training routines, whereas triathletes from the EG modified their running plans, but maintained their swimming and cycling routines. Testing was completed at week zero (pre) and week 5 (post) to monitor changes over the course of a 5-week training program. This would allow coaches and other professionals to have further knowledge about the effect of a low-volume and high-intensity running plan and the effects that this program had on athletic performance and muscular performance parameters.

Subjects

Sixteen male triathletes (age = 33 ± 5 years, age range = 24 to 42 years, body mass = 74 ± 5 kg, height = 176 ± 9 cm) volunteered for the study, which was performed according

to the ethical standards established by the Helsinki Declaration (2013) and was approved by the local ethical committee of the University of Jaen (Spain). Inclusion criteria were (a) older than 18 years, (b) actively competing in races, (c) a clean bill of health for the past 6 months, and (d) not engaged in a high-intensity training program. Three of the triathletes did not complete the study because of illness during the intervention period, and their data were excluded. Each signed a written informed consent before participation, completed a detailed questionnaire, and recorded race distances and times, training type, total distance, and training duration, which were confirmed by their respective coaches. The group competed in a sprint-distance triathlon race to validate the current performance status. Further information about participants—demographic and training background—is shown in Table 1.

Experimental Design

The study was conducted between March and April 2015. At the time of these observations, the triathletes had completed between 3 and 4 months of training. A parallel 2-group, longitudinal (pre and post) design was used. Thus, physical tests were performed before (pretest) and after (posttest) the 5-week intervention period. The triathletes were assigned and matched to 2 groups, EG and CG, based on their performance in a sprint-distance triathlon competition (overall race time). To investigate the effect of a HIIT-based running program, triathletes from the CG were asked to maintain their training routines, whereas triathletes from the EG modified their running plans, but maintained their swimming and cycling routines. Therefore, during the HIIT period, the CG performed continuous moderate-intensity training sessions (for swimming, cycling, and running), whereas the EG implemented their continuous moderate-intensity sessions for swimming and cycling and HIIT for running.

Training

All HIIT sessions performed within this training program have been investigated in previous studies on endurance

TABLE 1. Demographic and anthropometric data of the participants (mean, *SD*), and information about characteristics of their training routines and athletic performance.*

Variables	Whole group (<i>n</i> = 13)	CG (<i>n</i> = 6)	EG (<i>n</i> = 7)	<i>p</i>
Age (y)	33.8 (5.4)	31.8 (3.2)	35.4 (6.5)	0.181
Body mass (kg)	74.3 (5.8)	74.3 (7.2)	74.3 (5.6)	0.945
Height (m)	1.8 (0.09)	1.8 (0.07)	1.8 (0.06)	0.415
Training volume ($\text{h} \cdot \text{wk}^{-1}$)	15.9 (2.39)	15.2 (2.48)	16.5 (2.32)	0.339
Number of workouts (per wk)	9.8 (1.43)	9.8 (1.6)	9.7 (1.48)	0.888
Number of running sessions (per wk)	3.5 (0.77)	3.3 (0.81)	3.6 (0.78)	0.654
Running volume ($\text{km} \cdot \text{wk}^{-1}$)	31.9 (8.3)	30.0 (7.75)	33.6 (8.9)	0.463
Running volume ($\text{h} \cdot \text{wk}^{-1}$)	4.8 (1.29)	4.3 (1.0)	5.1 (1.26)	0.242
Overall race time (min)	64.9 (6.57)	65.2 (6.69)	64.9 (6.98)	0.912

*CG = control group; EG = experimental group.

athletes (1,8,31) and training volume and intensity were prescribed according to these works. Instructions for athletes regarding exercise intensity were given according to running pace, in terms of kilometers per hour (kmh). For this purpose, the variable used was the velocity associated with $\dot{V}O_{2\max}$ ($VV_{O_{2\max}}$) which was indirectly estimated through the velocity of a 3000-m race (11,18), information reported from the coaches.

The HIIT program included 3–4 sessions per week for 5 weeks. This running program led to a reduction of average weekly running distance in the EG (–69.8%, from 33.6 km per week before training program to 10.14 km per week during training program). A description of the 5-week HIIT-based program is reported in Table 2.

Materials and Testing

The triathletes were instructed to refrain from intense exercise two days preceding testing and to perform the last HIIT session 3 days before the posttest. They were not allowed to eat during the hour preceding the test or to consume coffee or other products containing caffeine during the preceding 3 hours. Pretesting and posttesting were conducted at the same time of day to avoid the influence of the circadian rhythm and under similar environmental conditions (20–24° C).

Either at pretest or posttest, participation involved the execution of a sprint-distance triathlon race (750-m swimming, 20-km cycling, and 5-km running), which was completely performed in simulated conditions, in the same sports facilities (closer than 100 m to each other). The triathlon involved swimming in an eight-lane, 25-m pool; and cycling on their own road racing bicycles, connected to the same electromagnetically braked roller (T2170; Tacx Vortex, Wassenaar, the Netherlands), which was calibrated to quantify and adjust wheel-ergometer rolling resistance to 1.1–1.6 kg, as prescribed by the manufacturer; and running on an outdoor 400-m synthetic track.

In both occasions (pretest and posttest), just before starting the race and after a standardized warm-up, the participants performed jumping tests (countermovement [CMJ] and squat jumps [SJ]) as baseline values. These measurements were repeated 3 more times, after swimming (Post-Sw), after cycling (Post-Cy), and after running (Post-Ru) to monitor the neuromuscular response during the competition. The participants were experienced athletes who performed jumping tests in their daily training sessions. Moreover, to make sure the execution was correct, a familiarization session was conducted during the last training session before testing. The CMJ and SJ were recorded using the OptoGait system (Microgate, Bolzano, Italy), which has been previously used in similar studies (21). Subjects performed two trials with a 15-second recovery period between them, and the best trial was used for the statistical analysis.

TABLE 2. Detailed description of the 5-wk high-intensity intermittent training-based running program, including exercises, intensity prescribed, and training volume.*

Exercise	Intensity (% $\dot{V}O_{2\max}$)	Week 1	Week 2	Week 3	Week 4	Week 5
100-m runs with 30-s rest	120–130	×20	×25	×30	×35	×20
400-m runs with 90-s rest between runs and 3-min between sets	105–110	2 × (4 × 400 m)	2 × (5 × 400 m)	2 × (5 × 400 m)	2 × (5 × 400 m)	2 × (3 × 400 m)
Bouts of 120-s runs with 120-s rest	100–105	×7	×7	×7	×7	–
30-s all-out running sprints with 3-min rest	All out	–	–	×4	×6	×6
Training volume	Session per wk km · wk ^{–1}	3 ~9.4	3 ~10.7	4 ~11.9	4 ~13.0	3 ~5.7

*~ means that training volume, in terms of kilometers per week, might vary according to the meters covered in some exercises.

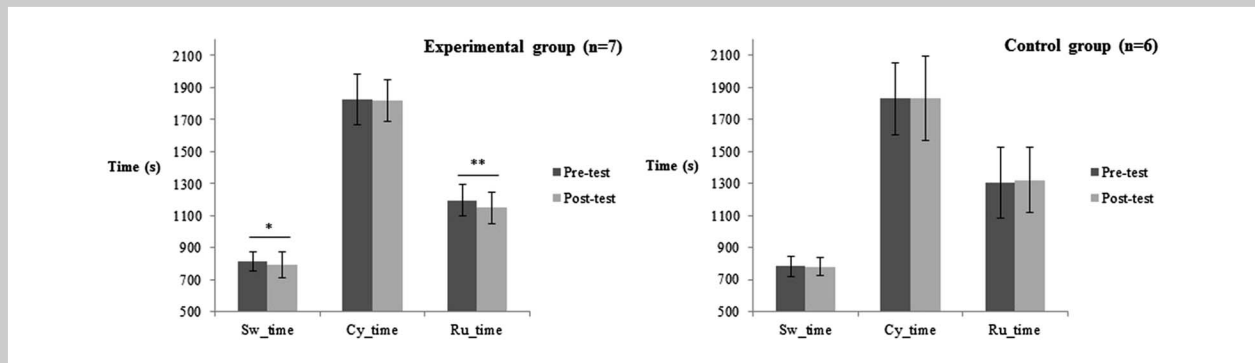


Figure 1. Athletic performance during a sprint-distance triathlon before (pretest) and after (posttest) a five-week HIIT-based training program. * $p \leq 0.05$; Sw_time: elapsed time for swimming; Cy_time: elapsed time for cycling; Ru_time: elapsed time for running.

The elapsed time (seconds) for the swimming, cycling, and running stages, and overall triathlon (transition times excluded) were registered for the subsequent analysis. Participants were experienced triathletes who had competed in these events; thus, the only instructions were to finish the race as fast as they could (transitions included). No other guidelines were provided as to exercise intensity, apart from the participants being informed that they were to exercise at an intensity of their own choice.

Additionally, to control the exhaustion level after the race and possible adaptations to the training program, some parameters were registered at both pretest and posttest: cardiovascular response—in terms of heart rate, in beats per minute—was monitored (Garmin Forerunner 405, Garmin International Inc., Olathe, KS, USA) throughout the

race, and the average heart rate (HR_{mean}) during every stage of the race was used for the analysis; the rate of perceived exertion (RPE) was recorded on the 6–20 Borg scale (5) immediately after the race; and blood lactate accumulation (BLA), at 1-minute after the race, was measured via fingertip blood samples, which were analyzed with a portable lactate analyzer (Scout Lactate; SensLab GmbH, Leipzig, Germany).

Statistical Analyses

Descriptive statistics are represented as mean (*SD*). Tests of normal distribution and homogeneity (Shapiro-Wilk and Levene's, respectively) were conducted on all data before analysis. Paired *t*-test was used to compare demographic data, body composition, and training background of

TABLE 3. Individual responses for the variables related to athletic performance (swimming, cycling, and running time, in addition to overall time) before (pretest) and after high-intensity intermittent training intervention (posttest).*

Group	Number of Participants	Pretest				Posttest			
		Sw_time (s)	Cy_time (s)	Ru_time (s)	Overall race time (s)	Sw_time (s)	Cy_time (s)	Ru_time (s)	Overall race time (s)
EG	1	729	2275	1150	4154	699	2212	1082	3993
EG	2	785	1895	1060	3740	724	1900	1030	3654
EG	3	826	2184	1179	4189	840	2178	1100	4118
EG	4	841	2171	1315	4327	840	2164	1290	4294
EG	5	990	1574	1206	3770	948	1571	1171	3690
EG	6	868	1745	1340	3953	835	1720	1270	3825
EG	7	676	1397	1123	3196	671	1380	1110	3161
CG	8	689	1791	1047	3527	693	1795	1094	3582
CG	9	789	1485	1222	3496	801	1492	1199	3492
CG	10	830	1980	1320	4130	821	1966	1352	4139
CG	11	823	1610	1649	4082	814	1624	1638	4076
CG	12	730	1686	1305	3721	727	1679	1322	3728
CG	13	843	2384	1295	4522	838	2404	1308	4550

*Sw_time = swimming time; Cy_time = cycling time; Ru_time = running time; EG = experimental group; CG = control group.

participants. A 2×2 analysis of variance (ANOVA) with repeated measures (group $\times$ measurement) was conducted for the dependent variables (time for swimming, cycling, running, and overall triathlon; and vertical jumping ability). The alpha was adjusted by Bonferroni correction. Nonparametric statistics were used with ordinal data—RPE—(Wilcoxon test, for within-group differences; Mann-Whitney U -test, for between-group comparison). Additionally, the magnitude of the differences between values was also interpreted using the Cohen's d effect size (ES) (30). Effect sizes of less than 0.4 represented a small magnitude of change, whereas 0.41–0.7 and greater than 0.7 represented moderate and large magnitudes of change, respectively (30). A Pearson correlation analysis was performed between the post-pre increase in elapsed time (Δ Sw_time, Δ Cy_time, Δ Ru_time, and Δ Overall_time, respectively), with the post-pre increase in baseline CMJ and SJ values (Δ CMJ and Δ SJ). Based on the findings from the correlation analysis, a simple linear regression analysis was used to predict Δ Ru_time and Δ Overall_time from the Δ CMJ during the intervention. The level of significance was $p \leq 0.05$. The data analysis was performed using SPSS (version 21; SPSS Inc., Chicago, IL, USA).

RESULTS

In a comparison between the CG and EG before the training program (pretest), no significant differences ($p \geq 0.05$) were found in demographic data or in body composition parameters. As for the characteristics of training routines and athletic level, the results obtained in both the CG and EG were similar with no significant differences ($p \geq 0.05$) (Table 1).

The results obtained regarding the effect of the training program on athletic performance are presented in Figure 1 and Table 3 (individual responses). Both the CG and EG showed similar athletic levels with no significant differences in swimming ($p = 0.511$), cycling ($p = 0.995$), or running ($p = 0.355$) before the intervention period (at pretest). However, according to 2×2 ANOVA, EG and CG swimming and running times were different ($p \leq 0.05$) after training program. Significant group-by-training interaction was found in the athletic performance after 5 weeks of training: the EG improved swimming time (2.90%, $p = 0.013$, ES = 0.438) and 5-km running performance (3.93%, $p = 0.001$, ES = 0.667), whereas the CG remained unchanged ($p \geq 0.05$, ES < 0.4).

Cardiovascular response during every stage and RPE and BLA after the race are shown in Table 4. No significant differences ($p \geq 0.05$) between groups were found at pretest or posttest. After the HIIT intervention, neither the CG nor the EG experienced significant alterations in any variable analyzed ($p \geq 0.05$, ES < 0.4).

The neuromuscular response to a sprint-distance triathlon race, measured by means of vertical jump ability (CMJ and SJ), and the effect of the five-week intervention period on

TABLE 4. Mean heart rate during every stage, and blood lactate accumulation and rate of perceived exertion after a sprint-distance triathlon race: before (pretest) and after (posttest) a 5-wk intervention period.*†

	CG (n = 6)			EG (n = 7)		
	Pretest	Posttest		Pretest	Posttest	
HRmean_Sw (bpm)	169.22 (11.6)	166.37 (13.0)	0.231	170.0 (10.1)	168.92 (11.2)	0.103
HRmean_Cy (bpm)	162.1 (7.65)	164.32 (8.2)	0.279	161.0 (5.43)	163.9 (7.0)	0.222
HRmean_Ru (bpm)	176.83 (10.0)	177.5 (8.21)	0.073	176.43 (12.6)	177 (9.6)	0.110
RPE (5–19)‡	16.5 (1.8)	17 (1.2)	0.322	15 (1.8)	15.5 (1.1)	0.330
BLA (mmol·kg ⁻¹ ·min ⁻¹)	8.87 (2.9)	8.6 (3.1)	0.089	11.5 (3.4)	11.49 (4.3)	0.122
						0.502
						0.501
						0.547
						0.194
						0.972

*CG = control group; EG = experimental group; HRmean_Sw = mean heart rate during the swimming stage; HRmean_Cy = mean heart rate during the cycling stage; HRmean_Ru = mean heart rate during the running stage; RPE = rating of perceived exertion; BLA = blood lactate accumulation.

†Between-group significant differences at pretest or at posttest.

‡Nonparametric tests were conducted with RPE—ordinal data.

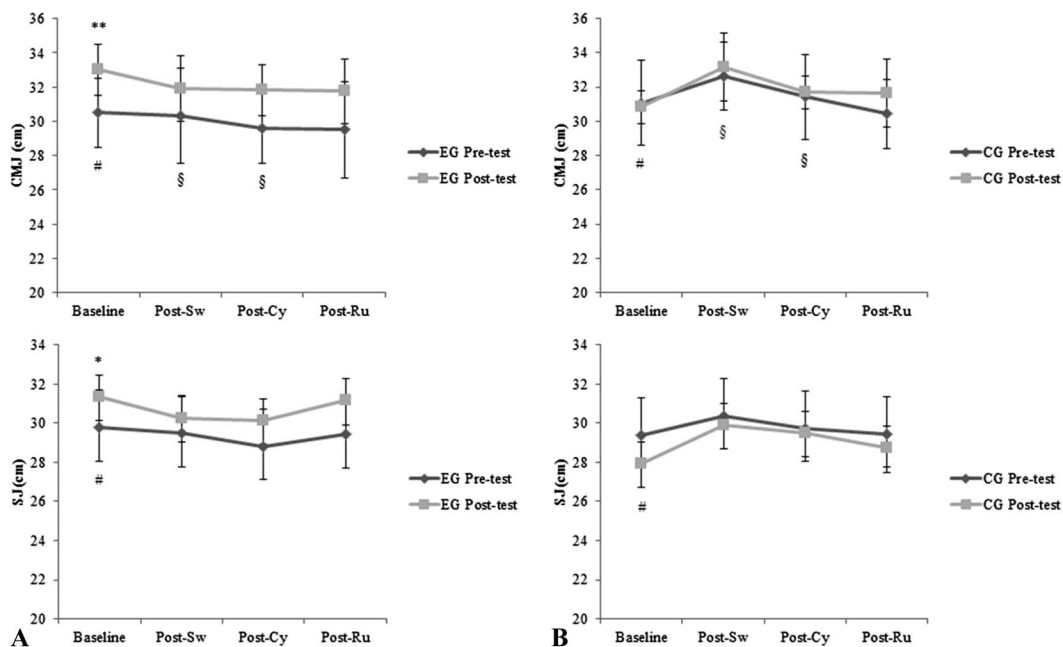


Figure 2. Neuromuscular performance, in terms of vertical jump ability, during a sprint-distance triathlon before (pretest) and after (posttest) a five-week HIIT-based training program for experimental (A) and CGs (B). * $p \leq 0.05$; ** $p < 0.01$; § indicates significant between-group differences ($p \leq 0.05$) at pretest; # indicates significant group-by-training interaction ($p \leq 0.05$). EG: experimental group; CG: control group; CMJ: countermovement jump; SJ: squat jump; post-Sw: measurement after the swimming stage; post-Cy: measurement after the cycling stage; post-Ru: measurement after the running stage.

this response are shown in Figure 2. Both the CG and EG showed similar performance before the HIIT intervention (CMJ: 30.5 and 30.98 cm; SJ: 29.77 and 29.43 cm, for the EG and CG, respectively; $p \geq 0.05$), but the performance was different after the training program ($p \leq 0.05$). Significant group-by-training interaction was found in vertical jumping ability: the EG improved CMJ (+9.21%, $p = 0.015$, ES = 1.498) and SJ performance (+5.9%, $p = 0.026$, ES = 1.065), whereas the CG experienced nonsignificant impairments in CMJ (−3.0%, $p = 0.373$, ES = 0.505) and SJ (−1.8%, $p = 0.228$, ES = 0.413). Concerning the dynamic of CMJ and SJ during the race, the repeated-measures analysis showed no significant changes ($p \geq 0.05$) at pretest or posttest for both the CG and EG.

The Pearson correlation analysis showed significant correlations between Δ CMJ and Δ SJ with Δ Ru_time ($p < 0.001$) and between Δ CMJ and Δ Overall_time ($p = 0.040$). A linear regression analysis showed that Δ CMJ predicted both the Δ Ru_time ($R = 0.748$; $R^2 = 0.559$; $p = 0.008$) and the Δ Overall_time ($R = 0.625$; $R^2 = 0.391$; $p = 0.048$).

DISCUSSION

The major finding of the present study was that the inclusion of a HIIT-based running plan with a reduction in training volume not only resulted in improved muscular performance (~6–9% in vertical jump ability) but also increased athletic

performance during a sprint-distance triathlon (improvements of 2.90% in swimming time, 0.47% in cycling time, and 3.93% in running time). Conversely, the triathletes from the CG, who continued their usual high-volume and low-/moderate-intensity training program, did not experience significant changes in muscular performance parameters or racing times. Additionally, the improvements reported by the EG in athletic performance were not accompanied by significant changes in the physiological response during the simulated race or in exhaustion level reached, which indicates that the triathletes experienced some adaptations that allow them to race faster with the same physiological impact.

To justify this study, some facts must be considered. First, a growing body of literature points to mean training intensity over a season as the key factor for performance improvement (8,20). It is also known that the risk of injury increases with increased weekly running distances in triathletes (9,28). With regard to this, the current running program led to a substantial reduction in average weekly running distance in the EG (−69.8%, from 33.6 km·wk^{−1} before the training program to 10.14 km·wk^{−1} during the plan), and an increase in average running pace (participants did not include HIIT sessions in their training routines before this training program).

Second, independent of the differences in distance and duration, all triathlons are considered continuous endurance

events (29). Despite the physiological basis of aerobic endurance being not clearly understood (6), it is well known that some of the major physiological determinants of endurance performance are work economy, lactate threshold, and maximal oxygen consumption (6). It has been shown that the presence of HIIT in endurance athletes' training programs facilitates the aforementioned adaptations (1,12,17,19). Likewise, the importance of high volumes performed at low/moderate intensity for maximizing athletic performance in endurance sports has also been demonstrated (22). For both reasons, a combination of high training volume at low exercise intensities and lower training volumes of HIIT seems to be necessary to obtain optimal development of endurance performance (7,8,20,27). In the current training program, all of these suggestions have been taken into consideration by ensuring the presence of low/moderate intensity over long periods of time (through swimming and cycling sessions) and by reducing weekly running volumes and increasing the average intensity of running sessions by means of HIIT.

Regarding the results obtained, this study is in agreement with previous works that have shown the effectiveness of HIIT programs for improving endurance performance and associated physiological variables (1,10,12,17,19). Focusing on athletic performance of trained individuals, the finding of a ~3–4% improvement in swimming and running performance after HIIT intervention is similar to the findings of Laursen et al. (19), who reported a 4.4% improvement in a 40-km cycling time trial after HIIT. Likewise, previous works (1,12) have reported an improvement in 3-km and 10-km running performance (3–7%) in endurance runners after different HIIT programs. To our knowledge, just 1 study (16) showed neither improvements nor decrements in athletic performance after a HIIT intervention in swimmers. As the authors explained, those results might be due to the extensive experience of the participants in HIIT exercises—criterion not met in the rest of studies. Previous studies performing HIIT interventions have been conducted in single sports, such as swimming (16), running (1,12,17), or cycling (19), but no previous work has proposed any strategy to insert and apply the HIIT methodology to a triathlon. In this regard, this study shows that the presence of HIIT in the triathletes' running plan not only improves running performance but also swimming performance during a sprint-distance competition, which might be associated with the “cross-training” principle (a phenomenon that refers to the cross-transfer of training effects from one sport to another) (24).

The precise mechanisms by which HIIT can improve endurance performance remain undetermined. Potential adaptations that may contribute to the improvement in endurance performance after HIIT include the increased ability of skeletal muscle to buffer hydrogen ions (32) and increased Na⁺/K⁺ pump capacity (1), anaerobic capacity (19), and motor unit activation (10,17). Although the acute neuromuscular, physiological, and metabolic responses were

not directly controlled in the present study, the data reported on the dynamic of HR, BLA, and muscular performance parameters during a sprint-distance triathlon before and after a 5-week HIIT plan let us gain some insight into the effectiveness of this training program. The results showed that a 5-week HIIT-based running plan improved vertical jumping capacity in triathletes, whereas triathletes who continued training at low-moderate intensities with high volumes (CG) did not experience changes in muscular performance variables. Additionally, the regression model performed in this study confirms the relationship between the gains in explosive muscular power and athletic performance improvements during a sprint triathlon. This finding supports the conclusion reported by Nummela et al. (25), who noted the importance of neuromuscular characteristics in determining running economy and, thereby, running performance. Likewise and regarding cycling performance, Faria et al. (14) indicated that peripheral adaptations in working muscles play a more important role for enhanced submaximal cycling capacity than central adaptations. And finally, with regard to the swimming performance improvement (although not statistically significant), the gains reported in explosive muscular power seem to maximize the positive effect of leg kick on the swimming speed—obvious direct generation of propulsive forces from the legs (15).

Hence, the improvements reported in this study highlight the effectiveness of a HIIT-based training program for improving explosive muscular power and accentuate the importance of neuromuscular performance in endurance performance. Nevertheless, the exact mechanism by which muscular performance improves after a period of HIIT in trained athletes is still unknown. What is clear is that a faster running pace during the HIIT sessions will demand higher levels of neural drive, will lead to higher levels of activation of the anaerobic glycolysis, and will recruit additional fast-twitch motor units for relatively short durations (26), which may be the physiological basis of improvements reported in this study.

Another important finding was the lack of changes in the physiological response during the simulated race at posttest, according to pretest data. The CG did not experience alterations in athletic performance and, thus, a similar cardiovascular response and BLA might be expected at pretest and posttest. However, the EG improved athletic performance at posttest, which was not accompanied by significant changes in the physiological response during the simulated race nor in exhaustion level reached, which indicates that the triathletes experienced some adaptations that allowed them to race faster with the same physiological impact. A right shift in the lactate threshold, so that higher running speeds are achieved at equivalent BLA levels, is a well-known consequence of endurance training and a determinant of endurance performance (23). Therefore, as well as muscle power improvements, physiological adaptations to the HIIT period might be determinants of athletic performance improvements reported in the current study.

CONCLUSIONS

The current study shows that a low-volume HIIT-based running plan combined with the already high training volumes of these triathletes in swimming and cycling is effective for improving athletic performance during a simulated sprint-distance triathlon competition. This improvement is suggested to be due to improved neuromuscular characteristics that were transferred into improved muscle power and work economy.

PRACTICAL APPLICATIONS

From a practical point of view, the current study offers insight into a training method prescription for triathletes by determining the effectiveness of a HIIT-based running program, which not only causes improvements in muscular performance parameters and allows athletes to enhance their athletic performance in a competition but also facilitates the reduction of a major risk factor for injury in triathletes such as weekly running distances.

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